Homework 6

Due: Friday, October 16 at 11:59 p.m. PT

MATH 112, Fall 2026

Instructions

- Write your full name, "Homework 6," and the date at the top of the first page.
- Show all work and explain your reasoning. Write in complete sentences.
- Typeset your solutions in LaTeX if you can.
- Submit a single PDF file to Gradescope under the assignment "Homework 6."
- Use Gradescope to electronically match problems to pages in your PDF.
- Questions? Email the instructor or come to office hours; see Moodle.
- You are strongly encouraged to work together. Write up your own solutions.

Be sure to show your work. Organize it so that another MATH 112 student who has not seen the problems can follow your reasoning from line to line. Solutions that contain only an answer, without justification or supporting work, will not receive full credit.

Problem 1 (5 points)

Prove that if $x, y \in \mathbb{R}$ with $x < y + \varepsilon$ for every $\varepsilon > 0$, then $x \leq y$.

Problem 2 (5 points)

Let A be a nonempty subset of $\mathbb{R}$, and suppose that $a_0 \in A$ has the property that $a \leq a_0$ for every $a \in A$. Prove that $a_0 = \sup A$.

In other words, if A has a maximum element, then that element is the supremum of A.

Problem 3 (15 points)

Let $a, b \in \mathbb{R}$ with $a < b$. Find the supremum and infimum of the set $(a, b) \cap \mathbb{Q} = \{x \in \mathbb{Q} : a < x < b\}$. Fully justify your answer.

Problem 4 (55 points)

Recall that $\mathbb{R}$ is complete, meaning that every nonempty subset of $\mathbb{R}$ that is upper bounded has a least upper bound. Given a subset $S \subseteq \mathbb{R}$, define $-S = \{-x : x \in S\}$.

(a) Suppose that $S \subseteq \mathbb{R}$ is nonempty and upper bounded, and let $M = \sup S$. Prove that $\inf(-S) = -M$.

(b) Suppose that $T \subseteq \mathbb{R}$ is nonempty and lower bounded. Prove that $\inf T$ exists in $\mathbb{R}$.
Hint: Use part (a) and the completeness of $\mathbb{R}$.

(c) **Infima without additive inverses.** The argument in part (b) uses the operation $x \mapsto -x$ on $\mathbb{R}$ to derive the existence of infima for nonempty lower bounded subsets. However, this holds generally. Let $(X, \leq)$ be a complete totally ordered set, i.e. it has the property that every nonempty subset of X that is upper bounded has a supremum in X. Prove that every nonempty subset of X that is lower bounded has an infimum.
Hint: Let $A \subseteq X$ be nonempty and lower bounded, and consider the set $L = \{x \in X : x \leq a \text{ for every } a \in A\}$ of all lower bounds of A. Apply the least-upper-bound property to L.

(d) Let $A, B \subseteq \mathbb{R}$ be nonempty bounded sets. Define $A + B = \{a + b : a \in A,\ b \in B\}$. Prove that

$$\sup(A + B) = \sup A + \sup B, \qquad \inf(A + B) = \inf A + \inf B.$$

For the infimum identity, you may use the relationship between suprema and infima established in part (a).

(e) Let $A \subseteq \mathbb{R}$ be nonempty and bounded, and let $t \in \mathbb{R}$. Define $tA = \{ta : a \in A\}$. Prove that:
(i) if $t \geq 0$, then $\sup(tA) = t \sup A$ and $\inf(tA) = t \inf A$;
(ii) if $t < 0$, then $\sup(tA) = t \inf A$ and $\inf(tA) = t \sup A$.
In particular, the case $t = -1$ recovers the relationship between S and $-S$ from part (a).

Problem 5 (20 points)

Write each of the following in the form $a + bi$, where $a, b \in \mathbb{R}$.

(a) $(1 + 2i)(3 - 2i) + (2 - 5i)$

(b) $(3 + 4i)(3 - 4i)$

(c) $\overline{4 - 6i}$

(d) $|3 - 4i|$

(e) $(3 + 2i)^2$

(f) $\mathrm{Im}\big(2+5i+i(3-7i)+17\big)$.
If $z = a + bi$ with $a, b \in \mathbb{R}$, the *imaginary part* of z is $\mathrm{Im}(z) := b$.

(g) $(1+i)^{-1}$

(h) $\dfrac{1-2i}{2+5i}$

(i) $(6+2i)\overline{(6+2i)}$

(j) $\dfrac{1}{i}$